DESCRIPTION OF COURSEWORK

Course Code	EEE406
Course Name	Power Systems Analysis
Lecturer	Dr Abdul Rashid Abdul Rahman
Academic Session	09/2024
Assessment Title	Laboratory 1 - Power flow studies using Gauss-Seidel and Newton-Raphson

A. Introduction/ Situation/ Background Information

Objectives

To perform power flow studies using Gauss-Seidel and Newton-Raphson for the given tasks.

Pre-Laboratory

Knowledge on power flow and familiarity with PowerWorld Simulator software is very important prior to conducting this laboratory. Therefore, students should complete the Pre-Laboratory steps below:

1. Study the power flow analysis through lecture notes or reference book.
2. Install the PowerWorld Simulator software.
3. Learn on how to use PowerWorld Simulator software from "PowerWorld Simulator Quick Guide Ver 5".

Watch related tutorial video at powerworld.com or youtube.com websites for better understanding on using PowerWorld Simulator software.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 1 Analyze transmission system and power flow with Gauss-Seidel and Newton-Raphson methods.

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
   ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.
   iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

This is a group of **four (4)** students's laboratory to be completed in 2 weeks.

Simulation files (PowerWorld's Simulator pwb and pwd files) should be submitted with the report using filename "Lab1ReportGroupX" in softcopy (PDF) for every task that should follow the following format:

1. Methodology
2. Results and Discussion

Due Date: **12th November 2024 (Week 9).**

Software Tool: PowerWorld Simulator software.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Design a reliable isolated network topology using logical slack bus voltage, generator schedule real power, generator voltage, impedance of the generators and lines/cables.	10
2.	Perform manual calculation using the system in part (1) to solve the voltages at all buses until FOUR (4) iterations using Gauss-Seidel method. Evaluate your results.	20
3.	Develop the system in part (1) in PowerWorld Simulator software and run the simulation using Gauss-Seidel method step by step until FOUR (4) iterations. Compare and discuss the manual calculation with simulation results for every iteration.	10
4.	Perform manual calculation using the system in part (1) to solve the voltages at all buses until TWO (2) iterations using Newton-Raphson method. Evaluate your results.	20
5.	Using the developed the system in part (3) in PowerWorld Simulator software, run the simulation using Newton-Raphson method step by step until TWO (2) iterations. Compare and discuss the manual calculation with simulation results for every iteration.	10
6.	Using the developed the system in part (3) in PowerWorld Simulator software, run the simulation using Newton-Raphson method until complete. Evaluate the overall power flow results to meet the required criteria in giving supply to the customers in that island. Now, run the simulation using Gauss-Seidel method until complete. Compare and analyze the accuracy, number of iterations and result convergence for both methods. Justify the suitable method to be adopted for this system in carrying out power flow study.	20
7.	Using the developed the system in part (3) in PowerWorld Simulator software, run the simulation until complete using Newton-Raphson method. Create a situation when any generator's reactive power output reaches its limit and discuss your observations and findings.	10
	TOTAL	**100**

F. Task(s)

Design a **reliable mesh or loop** isolated power system network. TABLE 1 provides the initial design parameters to be strictly adopted.

TABLE 1

Minimum total no. of bus	4 buses
Minimum no. of generator bus (PV bus)	2 buses
Base MVA	100 MVA
Voltage level	11 kV
Total real load	10 MW
Total reactive load	8 MVAR

Complying with the information given in **TABLE 1**, you are given flexibility to choose relevant and logical parameters to design your minimum 4-bus system. This flexibility allows you to decide on the network topology, number of generator bus (PV bus), number of load bus (PQ bus), slack bus voltage, generator schedule real power, generator voltage, load's real and reactive power, size and impedance of the generators as well as lines/cables. State all your assumptions where applicable.

1. Design a reliable isolated network topology using logical slack bus voltage, generator schedule real power, generator voltage, impedance of the generators and lines/cables.
2. Perform manual calculation using the system in **part (1)** to solve the voltages at all buses in per unit (up to four decimal places) until **FOUR (4)** iterations using Gauss-Seidel method. Evaluate your results.
3. Develop the system in **part (1)** in PowerWorld Simulator software and run the simulation using Gauss-Seidel method step by step until **FOUR (4)** iterations. Compare and discuss the manual calculation with simulation results for every iteration. Display the simulation snapshot results in your report. Submit the PowerWorld's Simulator software pwb and pwd files with filename "Lab1Task3".
4. Perform manual calculation using the system in **part (1)** to solve the voltages at all buses in per unit (up to four decimal places) until **TWO (2)** iterations using Newton-Raphson method. Evaluate your results.

5. Using the developed the system in **part (3)** in PowerWorld Simulator software, run the simulation using Newton-Raphson method step by step until **TWO (2)** iterations. Compare and discuss the manual calculation with simulation results for every iteration. Display the simulation snapshot results in your report. Submit the PowerWorld's Simulator software pwb and pwd files with filename "Lab1Task5".
6. Using the developed the system in **part (3)** in PowerWorld Simulator software, run the simulation using Newton-Raphson method until complete. Evaluate the overall power flow results to meet the required criteria in giving supply to the customers in that island. Now, run the simulation using Gauss-Seidel method until complete. Compare and analyze the accuracy, number of iterations and result convergence for both methods. Justify the suitable method to be adopted for this system in carrying out power flow study. Display the simulation snapshot results in your report. Submit the PowerWorld's Simulator software pwb and pwd files with filename "Lab1Task6a" and "Lab1Task6b" respectively.
7. In power flow theory, if a generator's reactive power output reaches its limit, then it is modeled as though it is a PQ bus. Using the developed the system in part (3) in PowerWorld Simulator software, run the simulation until complete using Newton-Raphson method. Create a situation when any generator's reactive power output reaches its limit and discuss your observations and findings. Display the simulation snapshot results in your report. Submit the PowerWorld's Simulator software pwb and pwd files with filename "Lab1Task7".

APPENDIX 1

MARKING RUBRICS

Component Title	Power flow studies using Gauss-Seidel and Newton-Raphson				Percentage (%)		
Criteria	Score and Descriptors					Weight (%)	Marks
	Excellent (5)	Good (4)	Average (3)	Need Improvement (2)	Poor (1)		
System design	100% reliable and unique isolated network topology design and using logical parameters of industry data	80% reliable and unique isolated network topology design and using logical parameters of industry data	60% reliable and unique isolated network topology design and using logical parameters of industry data	40% reliable and unique isolated network topology design and using logical parameters of industry data	plagiarized isolated network topology design	2	
Manual calculation to solve the voltages using Gauss-Seidel method	100% detail steps of calculation shown with comprehensively and correctly evaluated	80% detail steps of calculation shown with comprehensively and correctly evaluated	60% detail steps of calculation shown with comprehensively and correctly evaluated	40% detail steps of calculation shown with comprehensively and correctly evaluated	less than 40% detail steps of calculation shown with comprehensively and correctly evaluated	4	
PowerWorld Simulator simulation using Gauss-Seidel method	100% detail simulation conducted and compared with manual calculation for every iteration	80% detail simulation conducted and compared with manual calculation for every iteration	60% detail simulation conducted and compared with manual calculation for every iteration	40% detail simulation conducted and compared with manual calculation for every iteration	less than 40% detail simulation conducted and compared with manual calculation for every iteration	2	
Manual calculation to solve the voltages using Newton-Raphson method	100% detail steps of calculation shown with comprehensively and correctly evaluated	80% detail steps of calculation shown with comprehensively and correctly evaluated	60% detail steps of calculation shown with comprehensively and correctly evaluated	40% detail steps of calculation shown with comprehensively and correctly evaluated	less than 40% detail steps of calculation shown with comprehensively and correctly evaluated	4	
PowerWorld Simulator	100% detail simulation conducted	80% detail simulation conducted	60% detail simulation conducted	40% detail simulation conducted	less than 40% detail simulation	2	

simulation using Newton-Raphson method	**and compared with manual calculation for every iteration**	**and compared with manual calculation for every iteration**	**and compared with manual calculation for every iteration**	**and compared with manual calculation for every iteration**	**conducted and compared with manual calculation for every iteration**		
Overall power flow results discussion and comparison of Gauss-Seidel and Newton-Raphson methods	**100% detail overall power flow results being discussed with comparison of both Gauss-Seidel and Newton-Raphson methods and justification for method adoption**	**80% detail overall power flow results being discussed with comparison of both Gauss-Seidel and Newton-Raphson methods and justification for method adoption**	**60% detail overall power flow results being discussed with comparison of both Gauss-Seidel and Newton-Raphson methods and justification for method adoption**	**40% detail overall power flow results being discussed with comparison of both Gauss-Seidel and Newton-Raphson methods and justification for method adoption**	**less than 40% detail overall power flow results being discussed with comparison of both Gauss-Seidel and Newton-Raphson methods and justification for method adoption**	**4**	
Create a situation when any generator's reactive power output reaches its limit	**100% detail situation created for generator bus becomes load bus with observations and findings**	**80% detail situation created for generator bus becomes load bus with observations and findings**	**60% detail situation created for generator bus becomes load bus with observations and findings**	**40% detail situation created for generator bus becomes load bus with observations and findings**	**Less than 40% detail situation created for generator bus becomes load bus with observations and findings**	**2**	
					TOTAL		

Note to students: Please include the marking rubric when submitting your coursework.